

TEST STRATEGY DOCUMENT

Oil & Gas Domain – Web Applications

1. Objective

This document defines the overall testing strategy for web-based applications (portals and dashboards) in the Oil & Gas domain. It provides a scalable and reusable framework aligned with Agile (Scrum) delivery, ensuring high-quality releases with optimal test coverage and efficiency.

2. Scope

In Scope

- Web applications (portals, dashboards)
- Functional validation of business workflows
- System-level validation
- Regression cycles
- Data validation testing
- UAT support

Out of Scope

- Mobile applications
 - Performance and security testing (unless explicitly required)
 - Infrastructure and network-level testing
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3. Delivery Model

The testing strategy follows **Agile (Scrum)** methodology:

- Sprint-based iterative testing
 - Continuous integration of test activities within development cycles
 - Collaboration with Product Owners, Developers, and Business stakeholders
 - Early defect detection through shift-left practices
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4. Test Approach

4.1 Overall Approach

- Risk-based testing focusing on critical business flows (e.g., production monitoring, asset tracking)
- Shift-left testing for early validation
- Continuous regression testing across sprints
- Combination of manual and automated testing

4.2 Test Levels

Test Level	Description
Functional Testing	Validate business requirements and workflows
System Testing	End-to-end validation of integrated system
Regression Testing	Ensure no impact due to code changes
Data Validation	Validate data integrity across UI and backend
UAT Support	Assist business users during validation

5. Test Environment Strategy

Environment Details

Environment	Purpose
QA / Test	Primary testing environment for all test phases

Key Considerations

- Single environment increases dependency risks
 - Strict environment management and scheduling required
 - Controlled deployment cycles to avoid conflicts
 - Use of masked or synthetic test data
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6. Test Data Management

- Use of pre-defined datasets for repeatability
- Data validation against backend/database sources

- Data refresh strategy aligned with sprint cycles
- Handling of edge cases and negative scenarios

7. Automation Strategy

Automation Tools

- **Mabl** – Functional and regression automation
- **Playwright** – UI and API automation

Approach

- Focus on automating:
 - Critical business workflows
 - Regression scenarios
- API validation integrated with UI tests
- Automation execution aligned with CI/CD (where applicable)

Automation Coverage

Area	Coverage
UI Testing	Medium
API Testing	Medium
Regression Suite High priority	

8. Entry & Exit Criteria

Phase	Entry Criteria	Exit Criteria
Functional Testing	Requirements available, test cases ready	All critical defects closed
System Testing	Stable build deployed in QA	95% test cases passed
Regression Testing	Stable build available	No high severity defects open
UAT	Business scenarios defined	Business sign-off received

9. Defect Management Process

Defect Lifecycle

1. Defect Logging
2. Triage
3. Assignment
4. Fix
5. Retest
6. Closure

Severity Levels

Severity Description

Critical System crash, data loss

High Major functionality broken

Medium Partial functionality impacted

Low Cosmetic issues

10. Roles & Responsibilities

Role	Responsibilities
Test Lead	Strategy, planning, stakeholder communication
QA Engineers	Test design, execution, defect reporting
Automation Engineer	Automation script development & maintenance
Product Owner	Requirement clarification, UAT validation
Developers	Fix defects, support testing

11. Risk Management

Risk	Impact	Mitigation
Single QA environment	Testing delays	Strict scheduling & environment control

Limited automation coverage	Increased manual effort	Prioritize critical flows
Test data issues	Invalid test results	Controlled data management
Dependency on upstream systems	Delays in testing	Use mocks/stubs where possible

12. Test Deliverables

Phase	Deliverables
Planning	Test Strategy, Test Plan
Execution	Test Cases, Execution Reports
Closure	Test Summary Report, Metrics

13. Metrics & Reporting

Key Metrics

- Test case execution status
- Defect density
- Defect leakage
- Automation coverage
- Sprint-wise test progress

Reporting Frequency

- Daily status (during sprint)
 - Sprint summary report
 - Release-level summary
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14. UAT Support Strategy

- Provide business-ready test scenarios
- Assist users during execution
- Track and triage UAT defects

- Ensure closure before sign-off
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15. Assumptions & Dependencies

- Requirements are stable and well-defined
 - QA environment availability is consistent
 - Stakeholder availability for UAT
 - Timely defect fixes from development team
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16. Tools & Technologies

Category	Tools
Test Management	JIRA / Azure DevOps
Automation	Mabl, Playwright
Defect Tracking	JIRA
Data Validation	SQL / DB Tools

17. Approval

Role	Name	Signature	Date
Test Lead			